



The challenge of transactional and transformational leadership in projects

Ana K. Tyssen ^{a,1}, Andreas Wald ^{b,*}, Patrick Spieth ^{a,2}

^a *Strascheg Institute for Innovation and Entrepreneurship (SIIE), EBS Business School, EBS Universität für Wirtschaft und Recht, 65375 Oestrich-Winkel, Germany*

^b *European Business School Paris, 37/39 Boulevard Murat, 75016 Paris, France*

Received 6 December 2012; received in revised form 13 March 2013; accepted 28 May 2013

Available online xxxx

Abstract

Projects as a form of temporary organizing are different from standard organizational processes. Due to their temporary and unique nature, projects are characterized by discontinuous personal constellations and work contents. Although leadership research has called for a consideration of context factors and their effects on leadership, leadership in a temporary setting has hardly been investigated. We therefore extend transactional and transformational leadership theory by looking at it from the perspective of the temporary organization. We develop a research model with testable propositions on the effects of the temporary organizations' characteristics on leadership and on followers' commitment in projects.

© 2013 Published by Elsevier Ltd.

Keywords: Leadership; Temporary organizations; Project; Transformational leadership; Transactional leadership

1. Introduction

In leadership theory, transactional leadership and transformational leadership represent two complementary points of view. Transactional leadership focuses on the task-related exchange of actions and rewards between follower and leader. Conversely, transformational leadership emphasizes a person-orientation by aligning followers' needs with the organization's (higher) tasks and goals (Bass, 1990). Although several leadership approaches can generally be classified as either transactional or transformational, there are some that more explicitly deal with the decision of whether to focus on the tasks to be pursued or on the people to realize these tasks (Lussier and Achua, 2009). Most of the approaches share an underlying assumption that relatively stable leader–follower constellations are prevalent in permanent organizational structures. This is reflected by the notion of the “emergence” of leadership (Atwater et al., 1999), and the discussion of different temporal sequences in leadership research

(Bluedorn and Jaussi, 2008). Both imply a common history and future, as is the case in permanent organizations (Janowicz-Panjaitan et al., 2009). In contrast, a temporary exchange relationship does not include continuity of social relations, which serves as the basis for leader–follower relationships.

The use of temporary forms of organizing is increasing as organizations face growing uncertainties in dynamic environments (Hitt et al., 2007). Owing to these uncertainties, firms often choose flexible organizational structures. Organizational flexibility (Whittington et al., 1999) is realized by temporary organizations in the form of projects and programs (Ekstedt et al., 1999). The literature has identified several characteristics which distinguish temporary organizations from permanent organizations: limited duration, unique outcome, missing/ blurred hierarchies, higher uncertainty/risks, and heterogeneous work teams (Bakker, 2010; Brockhoff, 2006; Maaninen-Olsson and Müllern, 2009; Pich et al., 2002). In general, these characteristics pertain to all manifestations of temporary organizations, such as programs and projects, but in a particular case each of the characteristics can be more or less pronounced (Janowicz-Panjaitan et al., 2009).

Effective leadership plays an important role in ensuring the success of temporary organizations facing a high degree of

* Corresponding author. Tel.: +33 155 187777; fax: +33 155 187780.

E-mail addresses: ana.tyssen@de.bosch.com (A.K. Tyssen), andreaswald@ebs-paris.com (A. Wald), patrick.spieth@ebs.edu (P. Spieth).

¹ Tel.: +49 611 7102 1350.

² Tel.: +49 611 7102 1374.

uncertainty (Waldman et al., 2001). The important role of leadership and the specific characteristics of projects correspond to an increase in studies focusing on leadership in a project context. Whereas earlier studies mainly searched for factors that define successful project leaders (Zimmerer and Yasin, 1998), more recent research draws on more advanced leadership theories such as transformational leadership (Barber and Warn, 2005; Clarke, 2010; Keegan and Den Hartog, 2004; Strang, 2005, 2011) to study leadership in projects. With only few exceptions (Müller and Turner, 2007), existing studies focused on single project types (Keller, 2006; Strang, 2011) or industries/companies (Keegan and Den Hartog, 2004) and did not explicitly consider the effects of temporary organizations' characteristics and their varying degrees on leadership.

We assume that leadership behavior in temporary organizations causes effects different from those in permanent organizations and that existing leadership theory requires an extension. The aim of this article is therefore to extend transactional and transformational leadership theory by including the characteristics of temporary organizations. For this purpose, we develop a research model comprising propositions on the effects of the temporary organizations' characteristics on leadership and on its effectiveness. We focus on the project as the unit of analysis and consider transformational and transactional leadership behaviors of the project manager/leader. As transformational leadership is reported to have a particular positive effect on followers' commitment (Avolio et al., 2004) and commitment is supposed to form a link between leadership behavior and organizational performance (Steyrer et al., 2008), our model also includes propositions on leadership's effects on employees' commitment in projects.

The paper is organized as follows. In the next section we discuss the temporary organizations' characteristics before we elaborate on the implications of these characteristics for leadership. This is followed by Section 4 where we introduce transactional and transformational leadership and develop our research model. In Section 5, we discuss the contribution of our model in light of existing research on leadership in projects and we portray avenues for future research. The sixth section concludes the paper.

2. The rise of temporary organizations and their characteristics

While temporary organizations used to be the domain of project-based industries such as construction, film-making, and software engineering, a far-reaching projectification of organizational work can today be observed in almost every industry (Bakker, 2010; Sydow et al., 2004). Temporary organizations can be seen as aggregates of individuals temporarily collaborating for a shared cause. These undertakings usually represent some form of complexity in terms of roles, number, and interdependencies of participants and they have a predetermined duration after which the participants' constellation ceases to exist (Packendorff, 1995). These temporary organizations take the form of projects, programs, temporary teams, or task forces.

Given the large variety of manifestations of temporary organizations, their characteristics can be more or less pronounced in every single case (Janowicz-Panjaitan et al., 2009). For instance, all temporary organizations have a limited duration but there can be short-term projects with a comparatively low degree of complexity or complex long-term programs with a duration of several years. The research model we develop in this paper accounts for these differences and the propositions specify the effects of varying degrees of project characteristics on leadership.

Temporary organizations are often formed when a change is made to the organizational structures of a permanent organization (Whittington et al., 1999). This phenomenon reflects the overall dynamics of competition (Ekstedt et al., 1999), which defines the general conditions for leadership in temporary organizations. In addition to peripheral dynamics and uncertainties, project-inherent characteristics pose specific challenges to leadership (Chen et al., 2004): work within temporary organizations is described as a time-limited undertaking (Maaninen-Olsson and Müllern, 2009) that is unique in set up, tasks, and work contents (Brockhoff, 2006). Projects normally make use of non-routine processes, which differ from standard organizational processes, and therefore cause uncertainty (Pich et al., 2002), discontinuous personnel constellations (Parker and Skitmore, 2005) with heterogeneous backgrounds (Chen et al., 2004) and differing (hierarchical) roles outside the temporary organization (Baccarini, 1996; Packendorff, 1995). Table 1 summarizes the most important characteristics of temporary organizations.

Owing to the stated characteristics, effective leadership is crucial in temporary organizations, especially seeing as project members are often less committed than in permanent organizational settings (Keegan and Den Hartog, 2004). Zimmerer and Yasin (1998) identify poor or misdirected leadership to be one of the most important causes of project failure (Zimmerer and Yasin, 1998). For overall project team performance (Hoegl et al., 2004) and the realization of the projects' goals (Gällstedt, 2003), it is vital to establish commitment to project objectives, and to adhering to the project's values. Followers should also be motivated to engage in the actions needed to achieve the project goals. Therefore, leaders should evoke in followers the firm desire to be part of the project team (McDonough III, 2000). In the following section, we therefore elaborate on the

Table 1
Characteristics of temporary organizations.

Temporary organizations' characteristics	Implication
Limited and predefined duration	Short time horizon, hampers development of deeper social relations (e.g. trust)
Unique project outcome	Limited use of experiences and routines; leader must encourage and allow a certain degree of autonomy
Missing hierarchies	Authority gap
Higher uncertainty and risk involved	Low commitment of project members
Heterogenous work teams	Role ambiguity of project members

challenges leaders have to face with regard to the stated characteristics of temporary organizations.

3. The challenges of leadership in temporary organizations

For a while now, scholars have questioned the effectiveness of established leadership styles and approaches in time-limited and authority-limited settings (Cleland, 1967; Thamhain, 2004). However, little research has been undertaken in order to shed light on the underlying issues at play in such situations. Contemporary leadership theories describe leadership as a process of complex interactions between leaders and followers, focusing on relationships, interactions, and subjective perception (Yukl, 2001). Owing to a project's limited duration, the assumed exchange processes often cannot take place. Furthermore, a project's short-term orientation and the focus on immediate deliverables shorten team members' time-horizon (Marks et al., 2001). In contrast, companies' overall goals usually impede decisions and actions that require a longer time-horizon, such as investments in knowledge management systems or management control systems (Lindner and Wald, 2011; Love et al., 2005). Leaders must therefore find ways of including longer term aims in a short-termed project. According to some scholars, this difficult, yet vital leadership task includes creating a long-term vision (Christenson and Walker, 2004) and engendering subordinates' commitment (Herold et al., 2008) to the organizations' tasks. However, other scholars say that leadership in a given time frame merely calls for a focus on the actual tasks to be accomplished as time is too scarce to build relationships (Müller and Turner, 2007).

The outcome of a temporary undertaking is unique; therefore, the path that leads to this outcome is often marked by uncertainty (Atkinson et al., 2006). Although projects' degree of novelty varies and cannot generally be specified in character or extent (Brockhoff, 2006), they often need novel practices. Therefore, leaders in temporary organizations have to allow team members to display creativity. This implies that it is insufficient for leaders to merely give instructions (Goodman and Goodman, 1976).

In contrast to a permanent organization, a temporary undertaking is to some extent autonomous from its host organization's (line) structures and strategy (Sundstrom and DeMeuse, 1990). Owing to the fact that a variety of experts with vastly different backgrounds work together (Chiocchio and Essiembre, 2009), temporary undertakings normally operate with complex roles. Differing (hierarchical) roles outside the temporary organization (Baccarini, 1996; Packendorff, 1995) also cut across organizational boundaries (Engwall, 2003) and add to complexity and ambiguity. Moreover, people might collaborate coming from diverse educational backgrounds and different organizational units and cultures (Bell and Kozlowski, 2002). Therefore, a great potential for conflict arises (Jones and Deckro, 1993).

Most people working in permanent organizations rely on their supervisor in the permanent part of the organization, who is responsible for promotion, training, and reward. The leader of a temporary organization has therefore little *de facto* authority. He or she might not be able to display the full range of (hierarchical)

authority that is available in a permanent organization towards his or her subordinates. Certain forms of influencing, such as rewarding and punishing, are difficult to practice in the absence of hierarchical lines (Hodgetts, 1968). Of vital importance, especially as people from different organizational units have to collaborate, is commitment towards the common goals (Hoegl et al., 2004).

In summary, owing to the absence of a history of relationship building, leaders must base their ability to motivate project members, as well as to influence their perceptions, values and commitment, on behaviors other than rewarding and punishing. The challenges that are specific to temporary organizations pose limitations on leadership actions; at the same time, leadership in temporary organizations has to influence and facilitate a range of subordinates' behaviors.

4. Extension of leadership theory to incorporate temporary settings

In order to systematically analyze leadership in temporary organizations, we use the distinction of leadership approaches as being task- or people-oriented. While traditional task-oriented theories are predicated on rational processes, people-orientation theories focus on emotions and the values that are affected (Yukl, 1999). Respective leadership models can incorporate people, tasks, or a combination of both, situations and environmental characteristics, and the range of outcomes is based on a continuum (Tannenbaum and Schmidt, 1973). Elaborated models consider situational effects, and offer accentuated suggestions with combinations of task and people-orientations (House, 1971).

Transactional and transformational leadership behaviors reflect the dimensions of task and people orientation, and incorporate earlier leadership approaches. Bass and Riggio (2006) therefore described this approach as "full range leadership theory". This description has become a generally accepted leadership behavior distinction in research (Meindl, 1990). In contrast to Bass (1990), who claims that transactional and transformational leadership behaviors represent a new paradigm, Den Hartog et al. (1997) state that these theories incorporate many ideas from previous leadership research, sociology and political science. We follow this line of argumentation, and scrutinize the factors that constitute transactional and transformational leadership behavior in order to comprehensively pinpoint possible antecedents to temporary organizations' characteristics.

4.1. Transactional and transformational leadership in projects

Both transactional and transformational leadership consist of several factors, which relate to concepts in leadership and social research. For example, the transactional dimension focuses on tasks (Winkler, 2009), with factors that relate to path-goal theory and vertical dyad theory (Den Hartog et al., 1997). The underlying assumption of these theories is a series of leader–follower bargains and exchanges that provide the necessary motivation for the followers to pursue the path set. In contrast to this, the transformational leadership dimension encompasses

several person-oriented factors, such as charismatic leadership, neo-charismatic leadership, inspirational leadership and visionary leadership (Den Hartog et al., 1997; Winkler, 2009). The theory of transformational leadership assumes that a leader is able to bring about positive changes in followers' values, attitudes, perceptions, and expectations in a permanent organization (Bass, 1990; Yukl, 1999). Furthermore, a transformational leader focuses on people and their motivations, beliefs, and behaviors, and provides them with visions that satisfy their needs and desires (Lussier and Achua, 2009).

While the transactional–transformational concept can be seen on one continuum (Burns, 1978), several authors regard this concept on two separate dimensions. These authors state that a leader can display behaviors from both dimensions (Bryman, 1992). Since the characteristics of temporary organizations result in a broad range of task complexity, we follow Bass (1999) arguing that transformational leadership is part of transactional leadership, as it expands the range of leadership behaviors that have to be displayed.

Existing empirical studies provide a mixed picture of the effectiveness of transactional and transformational leadership behaviors in temporary organizations: Characteristics of transactional leadership behavior have been found to be beneficial in certain project types (Müller and Turner, 2010). However, transformational leadership has been shown to be one of the most effective leadership behaviors in times of change and uncertainty (Bass, 1990); therefore, this form of leadership seems of particular interest to temporary organizations that are the bearers of change for the permanent organization (Sydow et al., 2004). Transformational leaders have also been identified to have a strong, positive influence on subordinates' commitment, which is fundamental to the successful outcome of projects (Christenson and Walker, 2004).

In order to fully analyze the applicability of transactional and transformational leadership theories to the characteristics of temporary organizations, we will consider the respective factors that constitute transactional and transformational leadership in the following section. We will include an investigation of transformational and transactional leadership effectiveness in projects and present nine propositions.

4.2. Transactional leadership behavior, objectives, responsibilities and project duration

Transactional leaders provide rewards when followers do what is expected of them, and punish them when they do not behave as expected (Bass, 1990). Research results are discordant regarding the factors that constitute transactional leadership behavior (Bycio et al., 1995). Several researchers remark that the factors constituting transactional leadership, as formerly proposed by Bass (1985), are not reflected in empirical studies (Antonakis et al., 2003; Den Hartog et al., 1997). Taking these considerations and the characteristics of temporary organizations into account, we investigate two factors of transactional leadership that describe active exchanges between leaders and followers: An active manager is a

leader who monitors followers' work deviations and irregularities. In projects, active managers focus on planning projects, and structuring aspects and actions to ensure the projects' realization. As routines, processes and structures of permanent organizations are absent in temporary organizations, leaders use project management standards as substitutes and combine them with active planning and structuring (Hodgson, 2004). However, this implies that followers know what is expected of them, and that they clearly understand the messages and goals to be reached (Putnam and Sorenson, 1982). The *raison d'être* of temporary organizations is a non-routine goal, which in the ideal case is clearly defined and corresponds to a detailed project plan with given responsibilities. Therefore, transactional leadership seems to be particularly apt for these settings. Pawar and Eastman (1997) for example claim that leadership in times of relative goal stability is close to that of an administrative manager, rather than a visionary leader, as represented by transformational leadership. Placing these statements in the context of projects, transactional leadership behaviors seem a priori appropriate. We therefore propose that:

Proposition 1a. *Transactional leadership behavior will be especially effective in projects that have strong goal clarity rather than path-goal uncertainty.*

Proposition 1b. *Transactional leadership behavior will be especially effective in projects in which responsibilities have been defined clearly and made known to all followers.*

In line with this argument, contingent reinforcement as a second transactional factor denotes the reward of followers if the desired performance is attained (Den Hartog et al., 1997). This behavior does not need a deep relational history between the leader and follower, but can be displayed within time constraints and when there is a lack of trust and commitment. From this, we conclude that:

Proposition 2. *Transactional leadership will be especially effective in short project durations.*

4.3. Transformational leadership behavior, hierarchies, heterogeneity, and project duration

Transformational leaders use transcendental goals (Bass and Riggio, 2006), which encompass a range of factors in transformational leadership behavior. These factors have been confirmed by several researchers, who place different emphases on respective factors.

Charismatic leadership is seen by many researchers as the central factor of transformational leadership (Bass et al., 2003). Charisma enables a leader to transform follower needs and behaviors, and to provide a vision, as well as a sense of mission. This transformation requires that the followers identify with the leader, which separates charisma from another factor of transformational leadership, namely inspirational motivation (Den Hartog et al., 1997). The concept of charisma originated in the work of Weber (1947) who states that charismatic leadership is a means to satisfy the challenges of a new epoch of organizations that have overcome established traditional

forms by quickly assembling temporary work groups. So, established coordinative mechanisms have been replaced by teamwork, which needs stronger leader–subordinate relationships than the prior systems (Bass, 1990). Weber and others also state that charismatic leadership can have a negative connotation, as charismatic leaders can take advantage of followers' identification with them when their goals differ from those of the organization (Shamir et al., 1993). Also, researchers note that charismatic leadership deviates from transformational leadership (Conger, 1999) because it disregards the potential negative side of charismatic leaders (Weber, 1947). We agree with Bass (1985), who disregards the negative side of charismatic leadership by claiming charisma to be part of the transformational leadership behavior. Charismatic leaders lead groups successfully, irrespective of actual hierarchies, since their personalities and qualities are acknowledged to be extraordinary, and because they serve as role models (Bryman, 1992). Charismatic leaders thus advocate change, thereby challenging the status quo. Studies based on various neo-charismatic theories confirm that this leadership behavior leads to high levels of follower motivation and commitment, especially in times of uncertainty (House and Aditya, 1997). Consequently, charismatic leadership benefits projects which generally exist in uncertain conditions. Charismatic leadership hinges on identification with the leader, and seems to attain most success when hierarchies are absent and when the team constellations are heterogeneous and impermanent. We therefore propose:

Proposition 3. *Transformational leadership behavior is especially effective in projects that have missing or highly ambiguous hierarchies.*

Charismatic leaders emphasize giving a longer term vision and mission (Herold et al., 2008). Owing to projects' shorter termed tasks often pioneering longer termed goals (Christenson and Walker, 2004), these types of organizations are positively affected by charismatic leadership. Longer project durations also enable leaders and followers to establish and develop a more mature relationship. We thus propose the following:

Proposition 4. *Transformational leadership is especially effective in projects with longer duration.*

Intellectual stimulation as a further factor of transformational leadership refers to a leader's ability to elicit alternative approaches in followers, to solve a problem and to find new ways of dealing with challenges (Rafferty and Griffin, 2004). Intellectual stimulation fosters followers' creative and innovative behavior by encouraging them to question assumptions and reframe problems, and weigh up different approaches and solve problems actively (Winkler, 2009). This behavior suits projects and their unique outcomes, since the path to realize these outcomes is often marked by uncertainty (Atkinson et al., 2006). Even though the degree of project novelty varies and cannot generally be specified in character or extent (Brockhoff, 2006), temporary organizations generally need distinct and novel practices to effectively achieve the organizations final aims (Marks et al., 2001). This need for novel approaches indicates the beneficial leadership behavior of intellectual

stimulation in temporary settings, which supports project team members thinking creatively and acting innovatively (Uhl-Bien et al., 2007). In contrast, when leaders support intellectual stimulation, they allow subordinates to have some degree of independence (Morgeson, 2005). This, however, contrasts the leaders' liability for the projects progress, which rather causes the corresponding urge to stay in a position to make decisions exclusively. Regardless, leading as merely giving instructions to be followed is insufficient in temporary organizations (Goodman and Goodman, 1976). Owing to it being impossible to revert to routine processes and know-how, a leader should encourage creativity and learning (Pich et al., 2002), as well as autonomous decision-making (Heinz et al., 2006), all of which is encouraged by intellectual stimulation behavior. We thus suggest:

Proposition 5. *Transformational leadership is especially effective in projects with a high degree of novelty.*

Another factor constituting transformational leadership, inspirational motivation, is a leader's ability to articulate an attractive vision of a probable future, thereby providing followers with meaning (Keegan and Den Hartog, 2004). For this, followers do not have to identify with the leader, as would be the case with charismatic leadership. Rather, leaders who provide inspirational motivation establish team spirit and express higher values that are to be shared in a simple manner (Bass, 1990). The ability to establish a sense of team spirit seems to be of particular importance in temporary organizations. Owing to time limitations, there are restrictions on the evolvement of positive team features such as cohesiveness, trust, and shared values (Parker and Skitmore, 2005). Furthermore, team members come from diverse departments and locations (Maaninen-Olsson and Müllern, 2009) with differing values and beliefs (Jones and Deckro, 1993). Heterogeneity of team members in terms of different expertise could also result in a group of uncooperative individuals (Chiocchio and Essiembre, 2009). This highlights the importance of strengthening team spirit. Non-routine tasks to be accomplished furthermore require a leader who inspires by providing a vision (Christenson and Walker, 2004).

The same line of reasoning follows individualized consideration as the last factor of transformational leadership. This behavior indicates the leaders' ability to support and consider the follower's needs. The leader's role as a coach and an advisor is highlighted in this aspect of transformational leadership. A temporary team often consists of individuals with complementary skills who come from different departments (Zwikaël and Unger-Aviram, 2010). Therefore, individual consideration could play a crucial role in enhancing the commitment of individuals to the tasks to be accomplished (Emery and Barker, 2007). Paying special attention to each individual also fosters a two-way exchange and communication. This might substitute some of the absent hierarchies and the temporary nature of the collaboration (Cleland, 1967). Taking these considerations into account, we propose:

Table 2
Causalities between temporary organizations and leadership.

Temporary organizations' characteristics	Assumed effects on leadership
Limited and predefined duration	Transactional: + Transformational: –
Unique project outcome	Intellectual stimulation: +
Missing hierarchies	Individualized consideration + inspirational motivation: +
Higher uncertainty and risk involved	Idealized influence (charisma/vision): +
Heterogenous work teams	Inspirational motivation: +

Proposition 6. *Transformational leadership is especially displayed in projects with high member heterogeneity in terms of origin and skills.*

Table 2 summarizes the assumed coherences of projects and leadership.

4.4. Transformational leadership, transactional leadership and affective commitment

Research findings regarding the effects of transactional and transformational leadership behaviors in temporary organizations are ambiguous (Keegan and Den Hartog, 2004; Keller, 1992; Strang, 2005). This ambiguity is partly the result of different antecedents being used to scrutinize leadership behaviors, which could be influenced by other context factors. Even though project success is a relevant variable for most researchers, the concept is not clearly defined (Prabhakar, 2008). The absence of a clear definition makes a measurement even more difficult. Conversely, follower's commitment is a well-defined and crucial factor that leaders must influence in order to produce a positive project outcome in temporary settings (Hoegl et al., 2004). Commitment is also supposed to form a link between leadership behavior and organizational performance (Steyrer et al., 2008). We therefore analyze follower's commitment in the context of leadership behaviors.

Projects bring change to the overall organization (Sydow et al., 2004). This change can be small, for example in the case of a minor marketing project, or it can be huge, as organizational restructuring or the implementation of a new IT-system. In the latter cases, project participants are exposed to a large degree of change and uncertainty (Gällstedt, 2003), which imposes a high probability of low commitment (Armenakis et al., 1993). Notably, research on organizational behavior has shown that employee's affective commitment is crucial for an overall successful working outcome, as it denotes the individual's attitudes and loyalty towards the organization (McShane and Glinow, 2008). Affective commitment reflects the individual's emotional sense of belonging to an organization, and indicates the described coherences and positive effects of individuals' commitment to a given task or organization (McShane and Glinow, 2008; Meyer et al., 1993). Research has shown that a commitment to a particular change positively influences the willingness to actively participate in the change (Herold et al., 2007). Commitment to change is therefore a characteristic

needed to fulfill the tasks given. Other constructs, such as readiness for change and openness to change seem to be lesser indicators of the tasks' actual fulfillment (Herold et al., 2007). As projects inherit the notion of change (Lundin and Söderholm, 1995), these findings seem to be of particular interest for the identification of reliable predictors for employees' actions towards the successful fulfillment of project tasks. Research also indicates that transformational leadership has a positive impact on affective commitment (Bass, 1999). According to Bass (1995) employee commitment directly reflects whether transactional or transformational leadership is being used. We therefore conclude that:

Proposition 7a. *Both transactional and transformational leadership have a positive influence on followers' affective commitment to a project.*

Proposition 7b. *Transformational leadership behaviors show a stronger positive influence on followers' affective commitment than transactional leadership components.*

Projects not only face uncertainty due to environmental dynamics but also incorporate uncertainty themselves (Pich et al., 2002). Furthermore, projects are established in order to bring about novelty (Brockhoff, 2006) and change (Turner and Müller, 2003), and to circumvent barriers to organizational change (Sydow et al., 2004). Uncertainty is often the consequence of change (Bordia et al., 2004), and affects the whole organization. Within a project, uncertainty is caused by the work content (Baccarini, 1996). We therefore focus on the notions of uncertainty that further define the coherences of transactional and transformational leadership behaviors on commitment in projects.

4.5. Leadership in projects and task novelty (internal uncertainty)

We identified the lack of routines as a characteristic of temporary organizations. The extent of missing routines depends on the type of project, the degree of complexity, and the novelty of the outcome. Several scholars have studied the effect that the degree of task novelty has on leadership effectiveness in temporary organizations (Brockhoff, 2006; Heinz et al., 2006). Brockhoff (2006) states that the degree of novelty has a moderating effect on leadership in projects. Thus, we posit the following:

Proposition 8a. *Transformational leadership has a greater positive impact on followers' commitment in projects with a high degree of task novelty than in projects with a low degree of task novelty.*

Proposition 8b. *Transactional leadership has a greater positive impact on followers' commitment in projects with a low degree of task novelty than in projects with a high degree of task novelty.*

As forming projects is a response to dynamic environments (Ekstedt et al., 1999), uncertainty also applies to the environment within which the (permanent) parent organization operates. As such, we have to consider possible influences of this setting on temporary organizations.

4.6. Leadership in projects in times of environmental uncertainty

As transformational leadership is supposed to be particularly effective in times of uncertainty and change (Waldman et al., 2001), this not only encompasses the nature of tasks to be accomplished but includes the prevalent settings in which these actions take place (Atkinson et al., 2006). Here, the conditions within which a firm operates are seen as major factors of the employees' perception of dynamics and uncertainty in the industry in which the company operates (Gällstedt, 2003). As temporary organizations cause change, the resulting perception of change and uncertainty is even stronger in such an organization (Sydow et al., 2004). For this reason, perceived environmental uncertainty ought to be included when analyzing the effectiveness of leadership behavior in projects.

Milliken (1987) defines uncertainty as the individual's inability to precisely forecast something. This inability might be the result of a lack of information (Berger and Calabrese, 1975), or misleading and ambiguous information (Putnam and Sorenson, 1982). Whatever the reason might be, the distinctive feature of uncertainty is a sense of distrust in terms of future events' cause and effect relationships (DiFonzo and Bordia, 1998). Individuals therefore react to uncertainty by reducing or managing it. Environmental uncertainty amplifies the display of transformational leadership in temporary organizations. Research has also presented evidence for the negative consequences of individual's perceived uncertainty on commitment (Hui and Lee, 2000). We thus assume that:

Proposition 9a. *The higher followers' perception of environmental uncertainty is, the stronger the positive effects of transformational leadership will be on followers' project commitment.*

Proposition 9b. *The lower followers' perception of environmental uncertainty is, the lower the positive effects of transformational leadership will be on followers' project commitment.*

In summary, our propositions on the causes of followers' commitment in projects include different project characteristics, transactional and transformational leadership, and environmental uncertainty. Leadership effectiveness in projects is affected by projects' characteristics. The effects of these characteristics on leadership constitute the basis for the proposed causalities of transactional and transformational leadership. In addition, the *raison d'être* of projects has been regarded in particular. In this context, external uncertainty, caused by dynamic and fast pacing external environments serves as reason for the overall projectification of companies. Fig. 1 summarizes the proposition in a research model.

5. Discussion

In both permanent and temporary organizations, transformational leadership seems most effective during times of change. This form of leadership can positively affect followers' commitment. However, most of the assumed causalities in our research model only pertain to temporary organizations as they relate to characteristics of temporary organizations which pose specific challenges to leadership (Chen et al., 2004).

The contribution to knowledge of our model is twofold. First, for the discipline of project management and research on temporary organizations it lies in the identification of the implications of project characteristics for leadership. We complement existing research by accentuating that the characteristics of temporary organizations can be more or less pronounced in every single project and by specifying how these varying degrees affect leadership. On a more general level, our study also responds to a recurring call for a better theoretical foundation of project management (Cicmil and Hodgson, 2006; Hanisch and Wald, 2011; Söderlund, 2004). Second, for the discipline of leadership research we follow the call for considering more organizational context factors of leadership (Antonakis et al., 2003; Avolio et al., 2009; Porter and MacLaughlin, 2006). We complement transactional and transformational leadership theory by the perspective of temporary organizations which represent an important organizational context factor.

Conversely to permanent organizations, projects are characterized by a limited duration (Janowicz-Panjaitan et al., 2009). This restriction and its accompanying effects imply that transactional leadership is most effective. However, projects also operate with a potential lack of authority and high task novelty, and project managers seem only able to overcome this issue by means of transformational leadership. In addition, the pressure of internal and external uncertainty that projects face increases the need for transformational leadership. Generally, our considerations have shown that project characteristics partly amplify the need for transactional or transformational leadership, depending on which set of characteristics prevail.

5.1. Empirical test of the model

From a leadership behavior perspective, we have proposed that projects' characteristics and their accompanying uncertainty fundamentally affect leadership effectiveness, but we have not yet empirically confirmed these findings. We thus advocate the empirical testing of our propositions. In order to consider the different effects of projects' characteristics on leadership, and to add to knowledge on the range of possible effects and cross-influences of these characteristics, we propose a preliminary qualitative in-depth study as a first step. This study should include qualitative interviews with leaders and followers in projects. The interviews should allow insights in further causalities and coherences and for a refinement of the propositions. As a second step, the possible generalization of the results of the qualitative study could be attained in a quantitative, large-scale survey.

As other studies have investigated transactional and transformational leadership, the measurement used could well be considered in the context of the proposed inquiries. Researchers have empirically tested the effectiveness of transformational leadership in times of change and uncertainty (Waldman et al., 2001), which is reflected in projects. Furthermore, project characteristics have been operationalized by several researchers (Dvir et al., 1998; Price et al., 2002). Therefore, we propose that these scales be combined in order to empirically test our propositions.

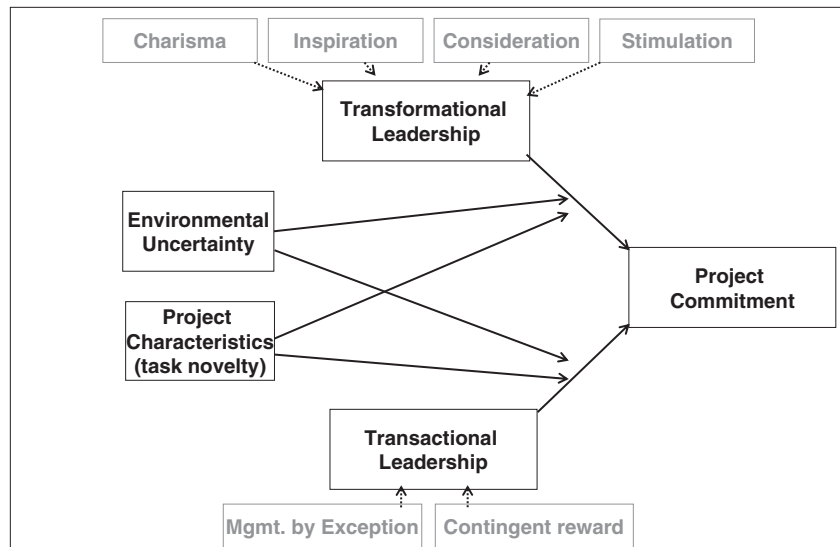


Fig. 1. Research model.

5.2. Future directions and extensions

Our propositions have major implications for leadership research. Most leadership theories are based on permanent organizations and their respective characteristics. Leadership research has to address the gaps caused by new forms of organizing. In this regard, several researchers have considered transformational leadership in temporary settings. For example, Barber and Warn (2005) conceptually link transactional and transformational leadership to projects, thus highlighting the need for project managers to use transformational leadership to proactively guide project members. The findings of Keegan and DenHartog (2004) indicate that the outcomes of transformational leadership in project settings tend to be weaker for employees reporting to project managers than for those reporting to line managers. In contrast, other scholars find that team members and sponsors negatively acknowledge a lack of transformational leadership. This passive or absent leadership negatively impacts project effectiveness and stakeholder satisfaction (Strang, 2005). Keller (2006) uses a longitudinal approach and reports that transformational leadership has a positive influence on project team outcomes. Overall, empirical investigations focus on individual projects or industries, producing mixed results. We believe that future empirical studies would benefit from using our model as a theoretical basis. Testing these as hypotheses on varying degrees of project characteristics could reduce the contradictory findings on the effects of transactional and transformational leadership in temporary settings.

Another research stream highlights the importance of inspiring and visionary project leadership (Christenson and Walker, 2004). Research combines elements of transformational leadership with emotional competences, as results indicate the importance of people-oriented leadership behaviors and provide evidence for different aspects of person-oriented behavior's positive influence on teams and task characteristics (Ayoko and Callan, 2010; Clarke, 2010; Müller and Turner,

2010). As other disciplines might provide insights in this regard, this also calls for a further incorporation of related research streams such as team research.

5.3. Implications for leadership practice

Our research has several implications for leadership practice. Along with the economic development, organizations and their structures are changing. As a result, leaders in organizations have to face new rationalities. Facing external as well as internal uncertainties, leaders have to anticipate their followers' fears. In general, the range and different set ups of temporary organizations leave it to the leader to decide which leadership behavior he or she should display. Our analysis has drawn attention to characteristics such as project duration and goal clarity, which call for the use of transactional leadership. This also highlights how important it is for project managers to underline path-goal clarity, as this seems to be a potential challenge in projects. In the broader context, leaders should emphasize how the temporary organization complements the parenting organizations' goals. Also, allowing room for discussion and individual decision making seems important in project teams with high task novelty, if project duration allows for this. Furthermore, the coherence between the single characteristics in respect to one another must be scrutinized. The question here is about which leadership style seems to be most effective in highly temporary teams with, for example heterogeneous experts, but a short overall duration.

5.4. Limitations

Our investigation has several limitations, which we would like to see addressed in further research. First, our conceptual analysis is restricted to one theoretical background. Further research might validate the theoretical findings on a different theoretical basis. Most research on leadership behavior in temporary organizations

places an emphasis on finding the leadership traits of successful project leaders. Several authors have combined different leadership research streams with regard to people-oriented dimensions, but they have neglected task-oriented leadership behaviors. We therefore propose to further scrutinize the existing leadership approaches with regard to the characteristics of temporary organizations.

Second, we propose that projects will be used more often as the new organizational rationale in many industries and services; therefore, we call for a broad testing across a range of industries and services in order to discover potential discrepancies.

6. Conclusion

The characteristics of temporary organizations seem indeed to impact leadership effectiveness in projects. We have made first attempts to extend leadership research in this direction. In this context, we have identified several challenges which leadership research and practice should consider when dealing with temporary organizations. In general, environmental uncertainty as an underlying rationale for the use of temporary organizations should be scrutinized in terms of its effects on leadership in projects. In particular, the transferability of predictions made by research on leadership in permanent organizations depends on the individual extent and ratio of project characteristics.

References

- Antonakis, J., Avolio, B.J., Sivasubramaniam, N., 2003. Context and leadership: an examination of the nine-factor full-range leadership theory using the Multifactor Leadership Questionnaire. *Leadership Quarterly* 14, 261–295.
- Amenakis, A.A., Harris, S.G., Mossholder, K.W., 1993. Creating readiness for organizational change. *Human Relations* 46, 681–703.
- Atkinson, R., Crawford, L., Ward, S., 2006. Fundamental uncertainties in projects and the scope of project management. *International Journal of Project Management* 24, 687–698.
- Atwater, L.E., Dionne, S.D., Avolio, B., Camobreco, J.E., Lau, A.W., 1999. A longitudinal study of the leadership development process: individual differences predicting leader effectiveness. *Human Relations* 52, 1543–1562.
- Avolio, B.J., Weichun, Z., Koh, W., Bhatia, P., 2004. Transformational leadership and organizational commitment: mediating role of psychological empowerment and moderating role of structural distance. *Journal of Organizational Behavior* 25, 951–968.
- Avolio, B.J., Walumbwa, F.O., Weber, T.J., 2009. Leadership: current theories, research, and future directions. *Annual Review of Psychology* 60, 421–449.
- Ayoko, O.B., Callan, V.J., 2010. Teams' reactions to conflict and teams' task and social outcomes: The moderating role of transformational and emotional leadership. *European Management Journal* 28, 220–235.
- Baccarini, D., 1996. The concept of project complexity — a review. *International Journal of Project Management* 14, 201–204.
- Bakker, R.M., 2010. Taking stock of temporary organizational forms: a systematic review and research agenda. *International Journal of Management Reviews* 12, 466–486.
- Barber, E., Warn, J., 2005. Leadership in project management: from firefighter to firelighter. *Management Decision* 43, 1032–1039.
- Bass, B.M., 1985. *Leadership and Performance Beyond Expectations*. Free Press, New York.
- Bass, B.M., 1990. *Bass & Stogdill's Handbook of Leadership — Theory, Research and Managerial Applications*, 3rd ed. Free Press, New York.
- Bass, B.M., 1995. Comment: transformational leadership: looking at other possible antecedents and consequences. *Journal of Management Inquiry* 4, 293–297.
- Bass, B.M., 1999. Two decades of research and development in transformational leadership. *European Journal of Work and Organizational Psychology* 8, 9–32.
- Bass, B.M., Riggio, R.E., 2006. *Transformational Leadership*, 2nd ed. Lawrence Erlbaum, Mahwah.
- Bass, B.M., Avolio, B.J., Jung, D.I., Berson, Y., 2003. Predicting unit performance by assessing transformational and transactional leadership. *Journal of Applied Psychology* 88, 207–218.
- Bell, B.S., Kozlowski, S.W.J., 2002. A typology of virtual teams: implications for effective leadership. *Group Organization Management* 27, 14–49.
- Berger, C.R., Calabrese, R.J., 1975. Some explorations in initial interaction and beyond: toward a developmental theory of interpersonal communication. *Human Communication Research* 1, 99–112.
- Bluedorn, A.C., Jaussi, K.S., 2008. Leaders, followers, and time. *Leadership Quarterly* 19, 654–668.
- Bordia, P., Hobman, E., Jones, E., Gallois, C., Callan, V.J., 2004. Uncertainty during organizational change: types, consequences, and management strategies. *Journal of Business and Psychology* 18, 507–532.
- Brockhoff, K., 2006. On the novelty dimension in project management. *Project Management Journal* 37, 26–36.
- Bryman, A.E., 1992. *Charisma and Leadership in Organizations*. Sage, London.
- Burns, J., 1978. *Leadership*. Harper & Row, New York.
- Bycio, P., Hackett, R.D., Allen, J.S., 1995. Further assessments of Bass's (1985) conceptualization of transactional and transformational leadership. *Journal of Applied Psychology* 80, 468–478.
- Chen, G., Donahue, L.M., Klimoski, R.J., 2004. Training undergraduates to work in organizational teams. *Academy of Management Learning and Education* 3, 27–40.
- Chiochio, F., Essiembre, H., 2009. Cohesion and performance: a meta-analytic review of disparities between project teams, production teams, and service teams. *Small Group Research* 40, 382–420.
- Christenson, D., Walker, D.H., 2004. Understanding the role of “vision” in project success. *Project Management Journal* 35, 39–52.
- Cicmil, S., Hodgson, D., 2006. New possibilities for project management theory: a critical engagement. *Project Management Journal* 37, 111–122.
- Clarke, N., 2010. Emotional intelligence and its relationship to transformational leadership and key project manager competences. *Project Management Journal* 41, 5–20.
- Cleland, D.I., 1967. Understanding project authority: concept changes manager's traditional role. *Business Horizons* 10, 63–70.
- Conger, J.A., 1999. Charismatic and transformational leadership in organizations: an insider's perspective on these developing streams of research. *Leadership Quarterly* 10, 145–179.
- Den Hartog, D.N.D., Van Muijen, J.J., Koopman, P.L., 1997. Transactional versus transformational leadership: an analysis of the MLQ. *Journal of Occupational and Organizational Psychology* 70, 19–34.
- DiFonzo, N., Bordia, P., 1998. A tale of two corporations: managing uncertainty during organizational change. *Human Resource Management* 37, 295–303.
- Dvir, D., Lipovetsky, S., Shenhar, A., Tishler, A., 1998. In search of project classification: a non-universal approach to project success factors. *Research Policy* 27, 915–935.
- Ekstedt, E., Söderholm, A., Wirdeus, H., 1999. *Neo-industrial Organising: Renewal by Action and Knowledge Formation in a Project-intensive Economy*. Routledge, London.
- Emery, C.R., Barker, K.J., 2007. The effect of transactional and transformational leadership styles on the organizational commitment and job satisfaction of customer contact personnel. *Journal of Organizational Culture, Communications & Conflict* 11, 77–90.
- Engwall, M., 2003. No project is an island: linking projects to history and context. *Research Policy* 32, 789–808.
- Gällstedt, M., 2003. Working conditions in projects: perceptions of stress and motivation among project team members and project managers. *International Journal of Project Management* 21, 449–455.
- Goodman, R.A., Goodman, L.P., 1976. Some management issues in temporary systems: a study of professional development and manpower — the theater case. *Administrative Science Quarterly* 21, 494–501.

- Hanisch, B., Wald, A., 2011. A project management research framework integrating multiple theoretical perspectives and influencing factors. *Project Management Journal* 42, 4–22.
- Heinz, U., Baga, T., Gebert, D., Kearney, E., 2006. Leadership and cooperation as success factors in innovative R&D projects on electronic platforms. *Team Performance Management* 12, 66–76.
- Herold, D.M., Fedor, D.B., Caldwell, S.D., 2007. Beyond change management: a multilevel investigation of contextual and personal influences on employees' commitment to change. *Journal of Applied Psychology* 92, 942–951.
- Herold, D.M., Fedor, D.B., Caldwell, S., Yi, L., 2008. The effects of transformational and change leadership on employees' commitment to a change: a multilevel study. *Journal of Applied Psychology* 93, 346–357.
- Hitt, M.A., Ireland, R.D., Hoskisson, R.E., 2007. *Strategic Management — Competitiveness and Globalization: Concepts and Cases*, 7th ed. South-Western, Mason.
- Hodgetts, R.M., 1968. Leadership techniques in the project organization. *Academy of Management Journal* 11, 211–219.
- Hodgson, D.E., 2004. Project work: the legacy of bureaucratic control in the post-bureaucratic organization. *Organization* 11, 81–100.
- Hoegl, M., Weinkauff, K., Gemuenden, H.G., 2004. Interteam coordination, project commitment, and teamwork in multiteam R&D projects: a longitudinal study. *Organization Science* 15, 38–55.
- House, R.J., 1971. A path goal theory of leader effectiveness. *Administrative Science Quarterly* 16, 321–339.
- House, R.J., Aditya, R.N., 1997. The social scientific study of leadership: quo vadis? *Journal of Management* 23, 409–465.
- Hui, C., Lee, C., 2000. Moderating effects of organization-based self-esteem on organizational uncertainty: employee response relationships. *Journal of Management* 26, 215–232.
- Janowicz-Panjaitan, M., Bakker, R.M., Kenis, P., 2009. Research on temporary organizations: the state of the art and distinct approaches toward “temporariness”. In: Kenis, P., Janowicz-Panjaitan, M., Chambre, B. (Eds.), *Temporary Organizations: Prevalence, Logic and Effectiveness*. Edward Elgar, Cheltenham, pp. 56–85.
- Jones, R.E., Deckro, R.F., 1993. The social psychology of project management conflict. *European Journal of Operational Research* 64, 216–228.
- Keegan, A.E., Den Hartog, D.N., 2004. Transformational leadership in a project-based environment: a comparative study of the leadership styles of project managers and line managers. *International Journal of Project Management* 22, 609–617.
- Keller, R.T., 1992. Transformational leadership and the performance of research and development project groups. *Journal of Management* 18, 489–501.
- Keller, R.T., 2006. Transformational leadership, initiating structure, and substitutes for leadership: a longitudinal study of research and development project team performance. *Journal of Applied Psychology* 91, 202–210.
- Lindner, F., Wald, A., 2011. Success factors of knowledge management in temporary organizations. *International Journal of Project Management* 29, 877–888.
- Love, P.E.D., Fong, P.S., Irani, Z., 2005. *Management of Knowledge in Project Environments*. Butterworth-Heinemann, Oxford.
- Lundin, R.A., Söderholm, A., 1995. A theory of the temporary organization. *Scandinavian Journal of Management* 11, 437–455.
- Lussier, R.N., Achua, C.F., 2009. *Leadership: Theory, Application, & Skill Development*, 2nd ed. South-Western, Eagan.
- Maaninen-Olsson, E., Müllem, T., 2009. A contextual understanding of projects — the importance of space and time. *Scandinavian Journal of Management* 25, 327–339.
- Marks, M.A., Mathieu, J.E., Zaccaro, S.J., 2001. A temporally based framework and taxonomy of team processes. *Academy of Management Review* 26, 356–376.
- McDonough, I.I., 2000. Investigation of factors contributing to the success of cross-functional teams. *Journal of Product Innovation Management* 17, 221–235.
- McShane, S., Glinow, M.V., 2008. *Organizational Behavior*, 4th ed. McGraw-Hill, New York.
- Meindl, J.R., 1990. On leadership: an alternative to the conventional wisdom. *Research in Organizational Behavior* 12, 159–203.
- Meyer, J.P., Allen, N.J., Smith, C.A., 1993. Commitment to organizations and occupations: extension and test of a three-component conceptualization. *Journal of Applied Psychology* 78, 538–551.
- Milliken, F.J., 1987. Three types of perceived uncertainty about the environment: state, effect, and response uncertainty. *Academy of Management Review* 12, 133–143.
- Morgeson, F.P., 2005. The external leadership of self-managing teams: intervening in the context of novel and disruptive events. *Journal of Applied Psychology* 90, 497–508.
- Müller, R., Turner, J.R., 2007. Matching the project manager's leadership style to project type. *International Journal of Project Management* 25, 21–32.
- Müller, R., Turner, R., 2010. Leadership competency profiles of successful project managers. *International Journal of Project Management* 28, 437–448.
- Packendorff, J., 1995. Inquiring into the temporary organization: new directions for project management research. *Scandinavian Journal of Management* 11, 319–333.
- Parker, S.K., Skitmore, M., 2005. Project management turnover: causes and effects on project performance. *International Journal of Project Management* 23, 205–214.
- Pawar, B.S., Eastman, K.K., 1997. The nature and implications of contextual influences on transformational leadership: a conceptual examination. *Academy of Management Review* 22, 80–109.
- Pich, M.T., Loch, C.H., De Meyer, A., 2002. On uncertainty, ambiguity, and complexity in project management. *Management Science* 48, 1008–1023.
- Porter, L.W., McLaughlin, G.B., 2006. Leadership and the organizational context: like the weather? *Leadership Quarterly* 17, 559–576.
- Prabhakar, G., 2008. What is project success: a literature review. *International Journal of Business and Management* 3, 3–10.
- Price, K.H., Harrison, D.A., Gavin, J.H., Florey, A.T., 2002. Time, teams, and task performance: changing effects of surface- and deep-level diversity on group functioning. *Academy of Management Journal* 45, 1029–1045.
- Putnam, L.L., Sorenson, R.L., 1982. Equivocal messages in organizations. *Human Communication Research* 8, 114–132.
- Rafferty, A.E., Griffin, M.A., 2004. Dimensions of transformational leadership: conceptual and empirical extensions. *Leadership Quarterly* 15, 329–354.
- Shamir, B., House, R.J., Arthur, M.B., 1993. The motivational effects of charismatic leadership: a self-concept based theory. *Organization Science* 4, 577–594.
- Söderlund, J., 2004. Building theories of project management: past research, questions for the future. *International Journal of Project Management* 22, 183–191.
- Steyrer, J., Schiffiger, M., Lang, R., 2008. Organizational commitment — a missing link between leadership behavior and organizational performance. *Scandinavian Journal of Management* 24, 364–374.
- Strang, K.D., 2005. Examining effective and ineffective transformational project leadership. *Team Performance Management* 11, 68–103.
- Strang, K.D., 2011. Eldership substitutes and personality impact on time and quality in virtual new product development projects. *Project Management Journal* 42, 73–90.
- Sundstrom, E., DeMeuse, K., 1990. Work teams: applications and effectiveness. *American Psychologist* 45, 120–133.
- Sydow, J., Lindkvist, L., DeFillippi, R., 2004. Project-based organizations, embeddedness and repositories of knowledge. *Organization Studies* 25, 1475–1489.
- Tannenbaum, R., Schmidt, W.H., 1973. How to choose a leadership pattern. *Harvard Business Review* 36, 162–170.
- Thamhain, H.J., 2004. Linkages of project environment to performance: lessons for team leadership. *International Journal of Project Management* 22, 533–544.
- Turner, J.R., Müller, R., 2003. On the nature of the project as a temporary organization. *International Journal of Project Management* 21, 1–8.
- Uhl-Bien, M., Marion, R., McKelvey, B., 2007. Complexity leadership theory: shifting leadership from the industrial age to the knowledge era. *Leadership Quarterly* 18, 298–318.
- Waldman, D.A., Ramirez, G.G., House, R.J., Puranam, P., 2001. Does leadership matter? CEO leadership attributes and profitability under conditions of perceived environmental uncertainty. *Academy of Management Journal* 44, 134–143.

- Weber, M., 1947. *The Theory of Social and Economic Organization*. Free Press, New York.
- Whittington, R., Pettigrew, A., Peck, S., Fenton, E., Conyon, M., 1999. Change and complementarities in the new competitive landscape: a European panel study, 1992–1996. *Organization Science* 10, 583–600.
- Winkler, I., 2009. *Contemporary Leadership Theories: Enhancing the Understanding of the Complexity, Subjectivity and Dynamic of Leadership*. Springer, Berlin.
- Yukl, G.A., 1999. An evaluation of conceptual weaknesses in transformational and charismatic leadership theories. *Leadership Quarterly* 10, 285–305.
- Yukl, G.A., 2001. *Leadership in Organizations*, 5th ed. Prentice Hall, Upper Saddle River.
- Zimmerer, T.W., Yasin, M.M., 1998. A leadership profile of American project managers. *Project Management Journal* 29, 31–38.
- Zwikael, O., Unger-Aviram, E., 2010. HRM in project groups: the effect of project duration on team development effectiveness. *International Journal of Project Management* 28, 413–421.